RAKESH RANJAN VEJENDLA

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EDUCATION

IU International University of Applied Sciences Oct 2024 – Nov 2026
M.Sc. Computer Science · Berlin, Germany

- Focus on machine learning, data engineering and software engineering; each module assessed by a delivered project.
- Thesis, completed November 2026: **an AI assistant for structured fault diagnosis on modified industrial machinery** — built from the machine's own documentation and fault records, it takes a technician from the symptom they can see to the part at fault and the setting it should be returned to; tested against real faults with known causes.

JNTUK University College of Engineering, Narasaraopet Aug 2018 – Sep 2022
B.Tech Mechanical Engineering · Andhra Pradesh, India

PROJECTS

Thyroid Disease Detection from Sleeping Heart Rate Python, LightGBM, scikit-learn

- Built a dataset of **5,568 cases from 175 patients** using the Garvan-UCI clinical archive, and engineered **8 features** from two sleeping heart-rate windows.
- The gradient-boosted model reached **91.5% sensitivity, 83.8% specificity and AUC 0.939** under patient-grouped cross-validation.

Scientific Image Forgery Detection Python, OpenCV, NumPy · Kaggle

- Built a training-free pipeline (SIFT, g2NN matching, RANSAC, dense re-check) that finds copied regions in a research figure in **~1.5 seconds on CPU**.
- Scored **0.375 mean overlap-F1 with zero false positives** on 60 authentic figures, against a 0.304 baseline that a deep-learning attempt had failed to beat.

ElderAssist — Care Marketplace React, TypeScript, PostgreSQL, Supabase

- Built a two-sided marketplace with two user roles, a full task lifecycle, and live updates pushed over realtime subscriptions.
- Enforced the access rules in the database: **row-level security on all six tables**, and a conditional update that keeps concurrent task acceptance safe.

Grocery Sales Batch Data Pipeline PySpark, Airflow, PostgreSQL, Docker

- Built an **11-task Airflow pipeline** that loads **5M+ sales rows** into a queryable warehouse, is safe to run twice, and stays within a 3.7 GB memory budget.
- Measured every performance claim, including two that failed: a Spark cluster ran **3.5× slower** than local execution, and JDBC partitioning gave no gain.

EXPERIENCE

upGrad Education Private Limited Feb 2023 – Aug 2023
Admissions Counsellor, Study Abroad · India

- Advised prospective students on international study programmes, explaining requirements, costs and timelines in plain terms.

Skill-Lync (EpowerX Learning Technologies) Nov 2022 – Feb 2023
Inside Sales Specialist · India

- Matched candidates to technical course tracks at an engineering upskilling company, based on their background and career goals.

SKILLS

Programming: Python, SQL, TypeScript / JavaScript

Machine learning & data: scikit-learn, LightGBM, PyTorch, pandas, NumPy, OpenCV

Data engineering: Apache Spark (PySpark), Apache Airflow, PostgreSQL, ETL pipeline design

Networking & systems: TCP/IP, DNS, HTTP/REST, wired and wireless connectivity, VPN fundamentals, Linux command line and system troubleshooting

Web & tools: FastAPI, React, Docker, Git, GitHub Actions, Linux

Languages: English (native), German (A2, B1 in progress), Telugu (native)